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EDUCATION

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- University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
• *B.S. - Computer Engineering (Autonomous Systems Focus); GPA: 3.99/4.00* *August 2022 - April 2026*
Courses: Robotic Control, Data Structures and Algorithms, Embedded Processors, Microelectronics

RESEARCH EXPERIENCE

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- Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
• *Undergraduate Robotacist Intern* *April 2023 - Present*
 - **ZOË 2 Rover:** (tinyurl.com/zoe2rover)
 - * Developing low-level software stack for the 2nd generation Zoë Rover set to conduct science in the Atacama Desert including CANopen protocols via the `ros_canopen` package for `ROS2` to communicate with encoders and motors.
 - * Conducted motor and motor controller datasheet specifications validation via physical testing.
 - **Underwater Snake Robot:** (tinyurl.com/humrsCMU)
 - * Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
 - * Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
 - * Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical maintenance.
 - **Apple's E-Waste Recycling Project:** (tinyurl.com/applecmu)
 - * Created datasets consisting of thousands of images for Machine Learning Models to detect screws in e-waste images.
 - * Integrated `ROS1` and `Python` packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
 - * Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

INDUSTRY EXPERIENCE

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- ESTAT Actuation** **Pittsburgh, PA**
• *Robotics Developer Co-op* *May 2025 - Present*
 - Building and setting up four new robotic test stands to improving testing capability by 133%.
 - Designing PCB for 3 different customer facing demos, making it easier to debug issues as they arrive.
 - Testing new generation rotary clutch prototypes, creating documentation explaining all the steps taken, and using these prototypes to inform design decisions for next generation production clutches.
 - Developing a PCB for a lightweight electroadhesive extendon system with potential applications in military and surgical uses.

STUDENT ORGANIZATIONS

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- Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
• *Chief Engineer/Integration Lead* *August 2022 - Present*
 - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and science teams. (tinyurl.com/roverimages)
 - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)
 - Securing and managing over \$7,000 in funding to support team development and future growth.
 - FIRST and VEX Robotics** **Exton & Royersford, PA**
• *Team Captain/Design Engineer* *August 2018 - June 2022*
 - Mentored younger students in Solidworks and manufacturing through workshops, enabling three-time Worlds qualification.
 - Won the VEX Judges Award, FIRST Excellence in Engineering Award, FIRST Industrial Design Award, and the FIRST Chairman's Award. (VEX Link: bit.ly/3w7a6b1) (FIRST Link: tinyurl.com/dwbot17)

PROJECTS

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- **555 Timer Roulette Board (KiCAD, Onshape, Circuit Design):** Designed and built a PCB-based LED Roulette Wheel using a 555 timer and CD4017 counter, featuring dynamic speed control through capacitive timing, dip switch power control, and a custom 3D-printed enclosure for portability. (tinyurl.com/555roulette)
 - **Bop It Project (KiCAD, Onshape, Microcontrollers):** Developed an interactive "Space Invaders Bop-It" reflex game using a bare ATmega328P chip, integrating analog inputs, audio-visual feedback, and custom embedded logic for randomized commands and progressively challenging gameplay. (tinyurl.com/spacebopit)
 - **Micromouse Project (Circuit Design, Robotic Control):** Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. (tinyurl.com/goyalmm)

- **Trains Project (Systems and Software Engineering):** (tinyurl.com/agtrain2025)
- **VoltVitals Embedded Board:** (tinyurl.com/voltvitals)
- **An Artistic take on Bipedal Robots (Onshape, Arduino):** Modified a preexisting bipedal robot to carry a symbolic lantern given to all freshmen at the University of Pittsburgh, reimagining a campus tradition through an artistic and technological lens for a humanities course on the intersection of art and technology. (tinyurl.com/gbipedal)
- **Calico AI Event Curator (AI, Website Development):** (tinyurl.com/steelcalicoai)
- **STM-32 Elevator Simulator (ARM Assembly, Project Integration):** Designed and Implemented software architecture in ARM Assembly to operate a physical elevator simulator PCB. (tinyurl.com/stmelevator)
- **Custom Cane (Human Centered Design, Solidworks):** Designed and fabricated a Walker-Cane Fusion to increase bathrooms accessibility for wheelchair users. Won first place at the Senior Design Expo. (tinyurl.com/CustomCane)
- **Formula SAE E-Brake Bias (Solidworks):** Developed an e-brake bias system for a formula style racecar, utilizing Solidworks and 3D printing technology to enhance performance and usability. Won the FSAE Innovation Award for the design and implementation of the project. (tinyurl.com/ebreakb)
- **Formula SAE Low-Cost Slip Angle Sensor (OpenCV, Raspberry Pi):** Worked to design and code prototypes of sensors which would allow for validation of slip angle using mouse sensors, digital cameras, and IMU's.
- **String Art Generator and Optimizer (Research, Python):** Developed an innovative string art optimization tools and GUI using Python programming to improve upon existing string art generators. (tinyurl.com/goyalstring)
- **Silicon Prosthetic Hand (High School Independent Research, Solidworks):** Designed and manufactured prototype prosthetic hand with silicone soft actuators and tested with human participants for an AP Research Project. (Link: <https://tinyurl.com/myprosthetic>)

SKILLS AND AWARDS

- **Languages:** Python, C++, C, ARM Assembly, RISC-V Assembly
- **Technologies:** Linux, ROS1, ROS2, Docker, Github, Solidworks, KiCAD, MATLAB, Microsoft Suite, Microcontrollers, OpenCV
- **Manufacturing:** Milling, Soldering, Laser Cutting, General Shop Tools
- **Awards:** Dean's Honor List (2021-Present), Honor List (2021-Present), Eagle Scout

PUBLICATIONS

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, A. Goyal, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, "**BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,**" in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)